CSE 4631

even odd decomposition:

$x[n] = \{1, 2, 3, 4\}$ (↑ 1)

$x[-n] = \{4, 3, 2, 1\}$ (↑ 1)

$$x_e = \frac{x[n] + x[-n]}{2}$$

$$x_o = \frac{x[n] - x[-n]}{2}$$

thus, for this example:

$\{0\ 0\ 0\ \underset{\uparrow}{1}\ 2\ 3\ 4\ 0\ 0\ 0\}$

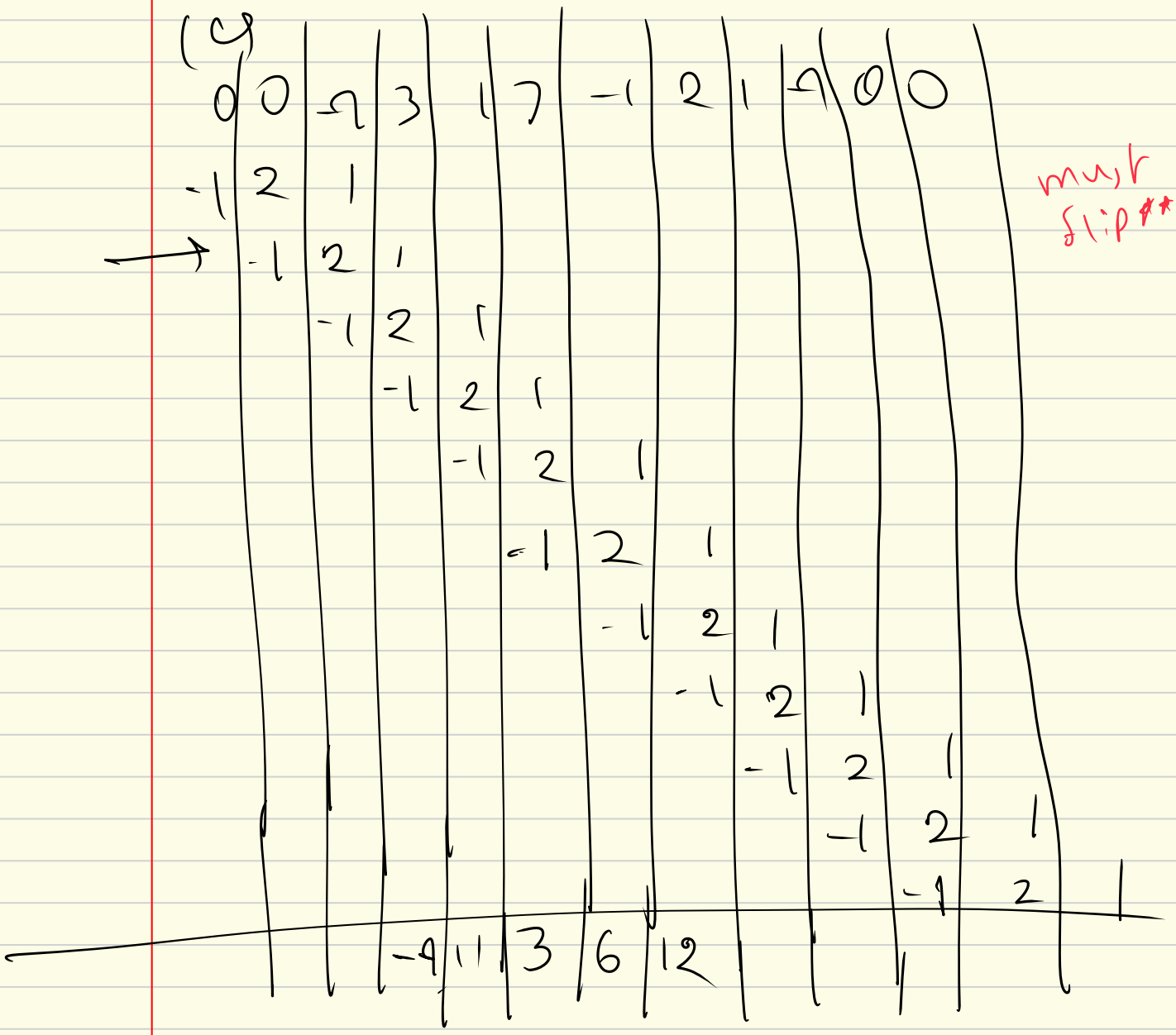
$\{4\ 3\ 2\ \underset{\uparrow}{1}\ 0\ 0\ 0\ 0\ 0\ 0\}$

(-3)(-2)(-1)(0)(1)(2)(3)(4)(5)(6)

$$\therefore x_e(-3) = \frac{0+4}{2} = 2$$

$$x_o(-3) = \frac{0-4}{2} = -2$$

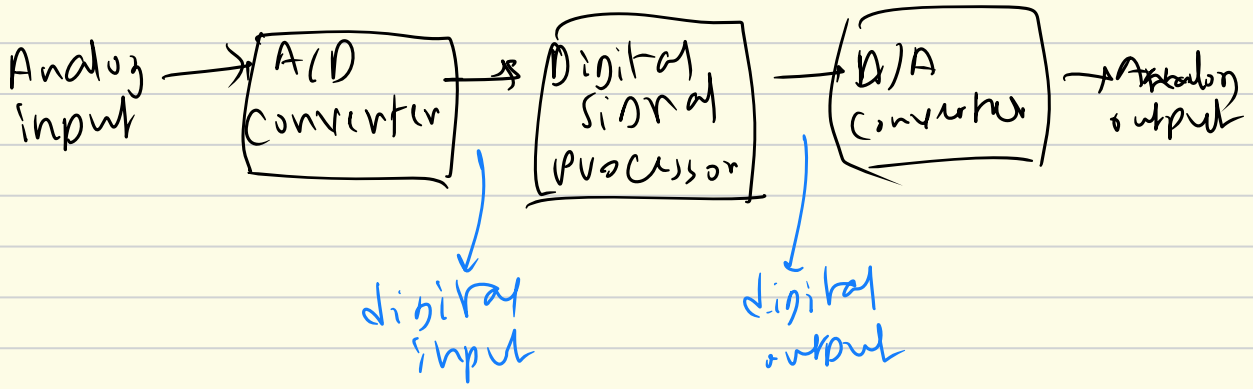
- for a system:
  - if the input is the delta function, output will be the impulse response
  - the amount of scaling and shifting done to delta function will be done to the impulse response as well
  - for different systems, impulse response will be different.


(c)
0 0 -2 3 1 7 -1 2 1 -2 0 0
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-1 2 1
-4 11 3 6 12
mult
flip**

Chapter 1:

Signal: A physical quantity that changes with time, space and/or any other variable.

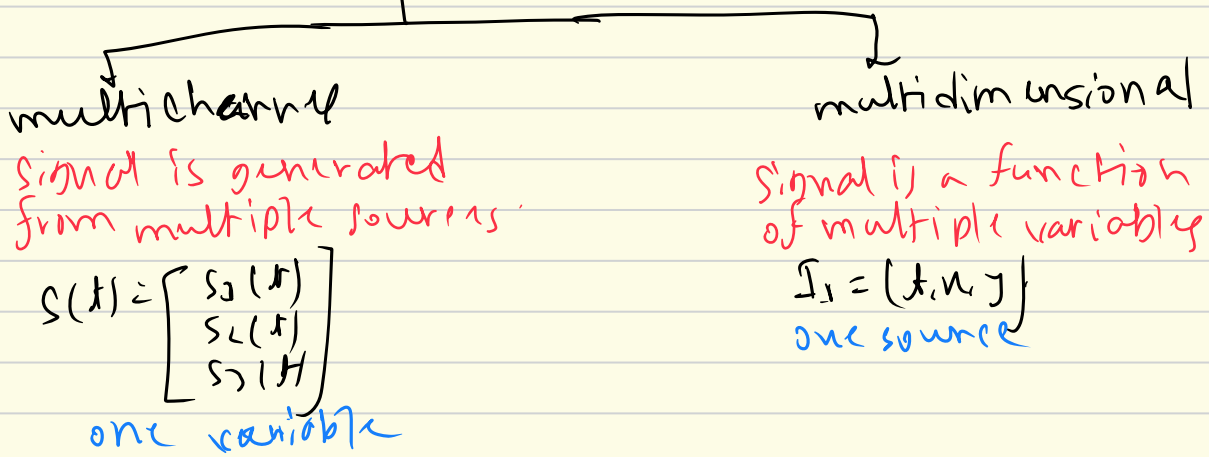
System: A physical device that performs an operation on a signal



Q: Why digital over signal?

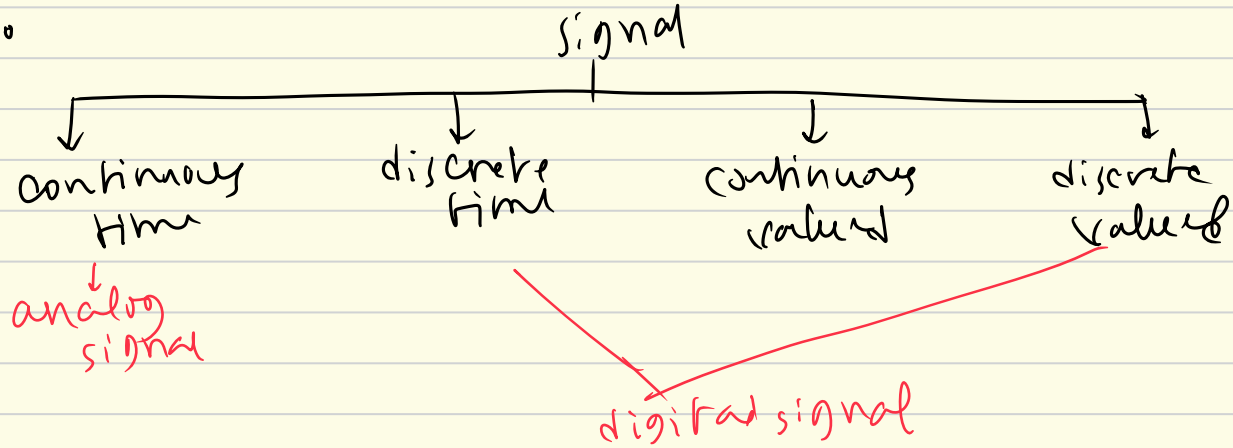
→ easier to reprogram a digital signal processor
→ higher accuracy operations
→ easier to store
→ easier to implement.

- Signal is a function of multiple variables, i.e. $S(t, v, \ldots)$



multichannel and multidimensional:

tv picture → $I(x, y, z) = \begin{bmatrix} I_0(x, y, z) \\ I_1(x, y, z) \\ I_2(x, y, z) \end{bmatrix}$



Deterministic: If a signal can be expressed using a mathematical expression, we call it deterministic.
All past, present, future values are known

Random: Not deterministic.

- continuous time sinusoidal:
$$x(t) = A\cos(2\pi f t + \theta)$$
discrete time sinusoidal:
$$x(n) = A\cos(2\pi f n + \theta)$$

Continuous time sinusoids:
(i) for all frequencies, it is periodic.
$$x_a(t+T) = x_a(t)$$
where $T = \frac{1}{f}$ is the fundamental period

(ii) sinusoidal signals with different frequencies are different (distinct)

(iii) Increasing frequency increases rate of oscillation as well.

$$x_a(t) = A\cos(\Omega t + \theta) = \frac{A}{2} e^{j(\Omega t + \theta)} + \frac{A}{2} e^{-j(\Omega t + \theta)}$$

to introduce negative frequency

Discrete time sinusoids:

(I) A sinusoidal signal is periodic if and only if the corresponding frequency is rational

proof:

If period is N, signal is periodic if:

$$x(n+N) = x(n)$$

$$\Rightarrow \cos[2\pi f_0(N+n)+\theta] = \cos[2\pi f_0 n+\theta]$$

$$\Rightarrow 2\pi f_0(N+n) = 2\pi f_0 n$$

This is only true if and only if

$$2\pi f_0 N = 2k\pi$$

$$\therefore \boxed{f_0 = \frac{k}{N}}$$

(II) Discrete time sinusoids whose frequencies are separated by an integer multiple of 2π are identical:

Proof: we need:

$$\cos[(\omega_0+2\pi)n+\theta] = \cos[\omega_0 n+\theta]$$

meaning:

$$x[n] = A\cos[\omega_k n+\theta] \text{ for all } k$$

where $\omega_k = \omega_0 + 2k\pi$ $\quad -\pi \le \omega_0 \le \pi$

Thus sinusoids with a frequency of $\omega_0 > \pi$ are identical to frequencies of $\omega_0 < \pi$

$\therefore \omega_0 > \pi$ is an alias of $\omega_0 < \pi$

[or range $0 \le \omega_0 \le 2\pi$]

(III) Highest rate of oscillation is achieved when $\omega = \pi$ (or $f = \frac{1}{2}$)

Proof: If we increase frequency, oscillation increases and period decreases

If $\omega_1 = \omega_0$ and $\omega_2 = 2\pi - \omega_0$

$\therefore$ For range $\pi \leq \omega_0 \leq 2\pi$

$x_1(n) = A\cos \omega_1 n = A\cos \omega_0 n$

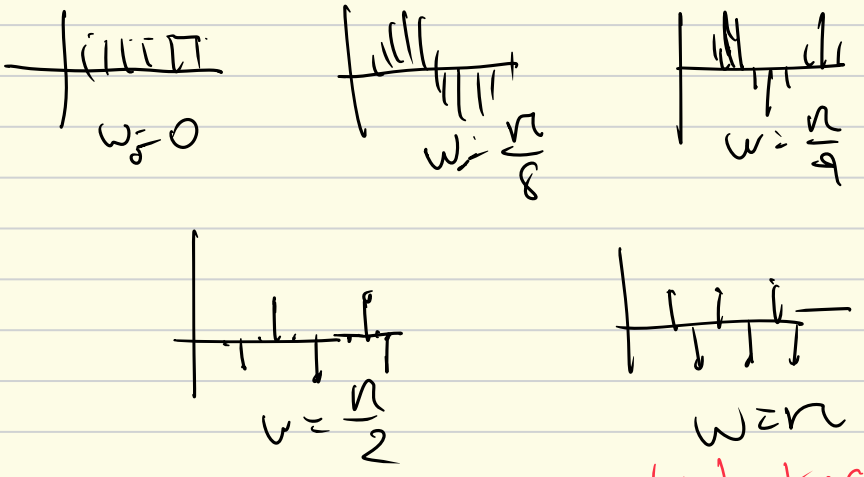
$x_2(n) = A\cos \omega_2 n = A\cos[(2\pi - \omega_0)n]$

$= A\cos(-\omega_0 n)$

$= A\cos\omega_0 n$

$= x_1(n)$

$\therefore$ Both sides are identical, centering around $\omega = \pi$



highest rate of oscillation.

A/D conversion:

(I) sampling: continuous time → discrete time

(II) Quantization: discrete time → discrete time
continuous valued → discrete valued

Each signal sample is assigned a discrete value from a finite set of discrete values

(IV) Encoding: Each discrete value is assigned

a b bit binary value.

Factors affecting accuracy in A/D conversion:
1. sampling rate
2. cost

- Difference between continuous time and discrete time signals is in the range of their frequency values

continuous: $-\infty < F < \infty$
$-\infty < \Omega < \infty$

discrete: $-\pi < w < \pi$
$-\frac{1}{2} < f < \frac{1}{2}$

Discrete time and continuous time relation:

$t = \frac{n}{F_s}$ discrete signal → continuous signal
ie: sample index to real time

or $f = \frac{F}{F_s}$ → continuous domain frequency (Hz)

normalized frequency (cycles)

used in discrete time version

or $w = \Omega T$

example: If we have 1kHz signal and $F_s = 8kHz$;
continuous time: $x = A\cos(2\pi \times 1000 \times t)$
discrete time: $x = A\cos\left(2\pi \frac{1000}{8000} \times t\right)$

$F_{max} = \frac{F_s}{2}$

$\Omega_{max} = \pi F_s$

$\frac{F_s}{2} \simeq \pi$

EXAMPLE 1.4.1

The implications of these frequency relations can be fully appreciated by considering the two analog sinusoidal signals

$$x_1(t) = \cos 2\pi(10)t$$
$$x_2(t) = \cos 2\pi(50)t \tag{1.4.12}$$

which are sampled at a rate $F_s = 40$ Hz. The corresponding discrete-time signals or sequences are

$$\begin{cases} x_1(n) = \cos 2\pi \left(\frac{10}{40}\right) n = \cos \frac{\pi}{2} n \\ x_2(n) = \cos 2\pi \left(\frac{50}{40}\right) n = \cos \frac{5\pi}{2} n \end{cases} \tag{1.4.13}$$

However, $\cos 5\pi n/2 = \cos(2\pi n + \pi n/2) = \cos \pi n/2$. Hence $x_2(n) = x_1(n)$. Thus the sinusoidal signals are identical and, consequently, indistinguishable. If we are given the sampled values generated by $\cos(\pi/2)n$, there is some ambiguity as to whether these sampled values correspond to $x_1(t)$ or $x_2(t)$. Since $x_2(t)$ yields exactly the same values as $x_1(t)$ when the two are sampled at $F_s = 40$ samples per second, we say that the frequency $F_2 = 50$ Hz is an *alias* of the frequency $F_1 = 10$ Hz at the sampling rate of 40 samples per second.

- $F_k = F_0 + F_s \times k$ is indistinguishable from F_0, thus, these frequencies are called aliases of F_0

As: $F_k = F_0 + F_s \times k$

$$\therefore x(n) = A\cos\left(2\pi \frac{F_0 + F_s k}{F_s} t + \theta\right)$$
$$= A\cos(2\pi F_0 t + 2\pi k t + \theta)$$
$$= A\cos(2\pi F_0 t)$$

Any frequency above $\omega = \pi$ can be mapped to a frequency in $0 \le \omega \le \pi$. Thus, we can use $\omega = \pi$ or $f = \frac{1}{2}$ to fold the frequencies. this is called the pivot point or the folding frequency.

- Interpolation function that is used to recover the original signal from the sample values is:

$$g(t) = \frac{\sin 2\pi B t}{2\pi B t}$$

- Half of the sampling frequency is what we can represent

EXAMPLE 1.4.4

Consider the analog signal

$$x_a(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12{,}000\pi t$$

(a) What is the Nyquist rate for this signal?

(b) Assume now that we sample this signal using a sampling rate $F_s = 5000$ samples/s. What is the discrete-time signal obtained after sampling?

(c) What is the analog signal $y_a(t)$ that we can reconstruct from the samples if we use ideal interpolation?

Solution.

(a) The frequencies existing in the analog signal are

$$F_1 = 1 \text{ kHz}, \qquad F_2 = 3 \text{ kHz}, \qquad F_3 = 6 \text{ kHz}$$

Thus $F_{\max} = 6$ kHz, and according to the sampling theorem,

$$F_s > 2F_{\max} = 12 \text{ kHz}$$

The Nyquist rate is

$$F_N = 12 \text{ kHz}$$

(b) Since we have chosen $F_s = 5$ kHz, the folding frequency is

$$\frac{F_s}{2} = 2.5 \text{ kHz}$$

and this is the maximum frequency that can be represented uniquely by the sampled signal. By making use of (1.4.2) we obtain

$$x(n) = x_a(nT) = x_a\left(\frac{n}{F_s}\right)$$

3 and 5 are greater than 2.5, so they're folded

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(\frac{3}{5}\right)n + 10\cos 2\pi\left(\frac{6}{5}\right)n$$

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(1-\frac{2}{5}\right)n + 10\cos 2\pi\left(1+\frac{1}{5}\right)n$$

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(-\frac{2}{5}\right)n + 10\cos 2\pi\left(\frac{1}{5}\right)n$$

Finally, we obtain

$$x(n) = 13\cos 2\pi\left(\frac{1}{5}\right)n - 5\sin 2\pi\left(\frac{2}{5}\right)n$$

The same result can be obtained using Fig. 1.4.4. Indeed, since $F_s = 5$ kHz, the folding frequency is $F_s/2 = 2.5$ kHz. This is the maximum frequency that can be represented uniquely by the sampled signal. From (1.4.17) we have $F_0 = F_k - kF_s$. Thus F_0 can be obtained by subtracting from F_k an integer multiple of F_s such that $-F_s/2 \le F_0 \le F_s/2$. The frequency F_1 is less than $F_s/2$ and thus it is not affected by aliasing. However, the other two frequencies are above the folding frequency and they will be changed by the aliasing effect. Indeed,

$$F_2' = F_2 - F_s = -2 \text{ kHz}$$

$$F_3' = F_3 - F_s = 1 \text{ kHz}$$

From (1.4.5) it follows that $f_1 = \frac{1}{5}$, $f_2 = -\frac{2}{5}$, and $f_3 = \frac{1}{5}$, which are in agreement with the result above.

(c) Since the frequency components at only 1 kHz and 2 kHz are present in the sampled signal, the analog signal we can recover is

multiply with 2 and convert to Hz

$$y_a(t) = 13\cos 2000\pi t - 5\sin 4000\pi t$$

which is obviously different from the original signal $x_a(t)$. This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.

Quantization:

quantization error

$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2}$$

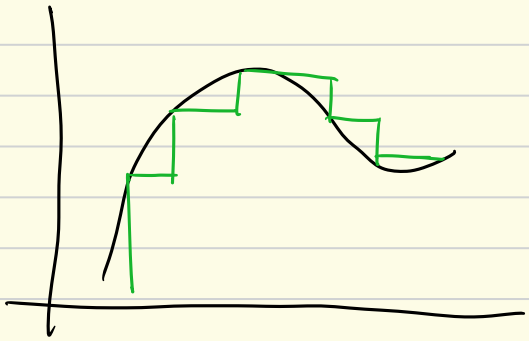
$$\Delta = \frac{x_{max} - x_{min}}{L-1}$$

Δ: quantization step

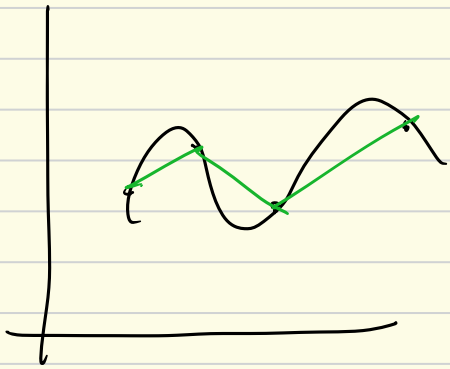
with b bits and L levels:

$$2^b \geq L$$

D/A converter:



zero order: value of one sample is held until the next one is received



linear interpolation: successive samples are connected with straight line segments.

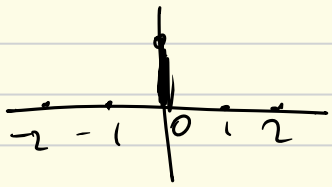
Chapter 2:

- Discrete time signal representations:
  - graphical representations
  - functional
  - tabular
  - sequence

- Elementary discrete time signal:

① unit sample:

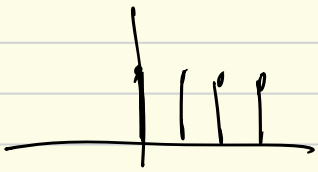
$$\delta(n) \equiv \begin{cases} 1 ; & n = 0 \\ 0 ; & n \neq 0 \end{cases}$$



also called unit impulse

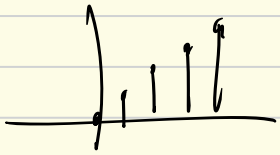
② unit step signal:

$$u(n) \equiv \begin{cases} 1 ; & n \geq 0 \\ 0 ; & n < 0 \end{cases}$$



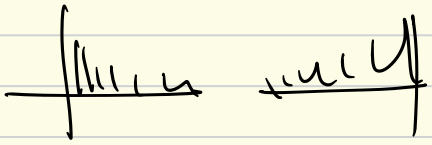
③ unit ramp signal:

$$u(n) \equiv \begin{cases} n ; & n \geq 0 \\ 0 ; & n < 0 \end{cases}$$



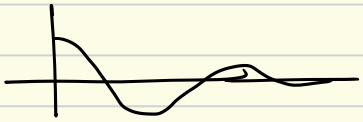
④ exponential:

$$x(n) = a^n$$
$$= r^n e^{j\theta n}$$
$$= r^n (\cos\theta n + j \sin\theta n)$$



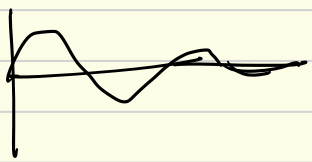
real:

$$x(n) = r^n \cos\theta n$$



imaginary:

$$x(n) = r^n \sin\theta n$$



Energy and power signals:

energy signal ⇒ energy is finite
- ⇒ decays over time
- → sum of squared magnitudes
- → calculated over entire lifetime

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

power signal ⇒ power is finite and nonzero, energy is infinite
- ⇒ average of energy over time
- ⇒ calculates the impact at any one moment

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

Q// folding and time delaying are not commutative?

$TD[x(n)] = x(n-k)$ (shift right)

$FD[x(n)] = x(-n)$

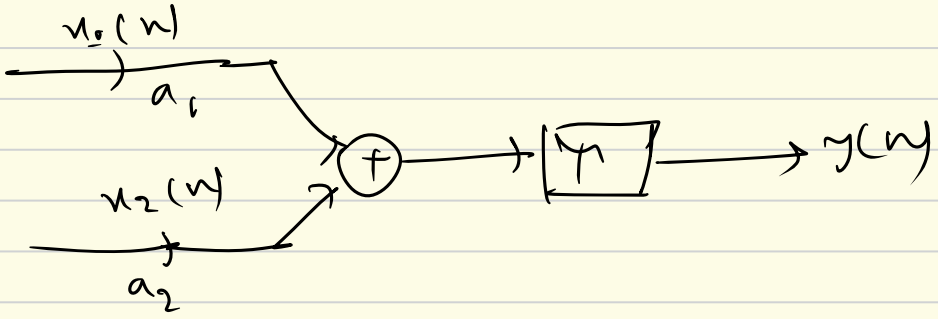
$\therefore TD[FD(n)] = TD[x(-n)] = TD[x(-n+k)]$

$FD[TD(n)] = FD[x(n-k)] = TD[-n-k]$

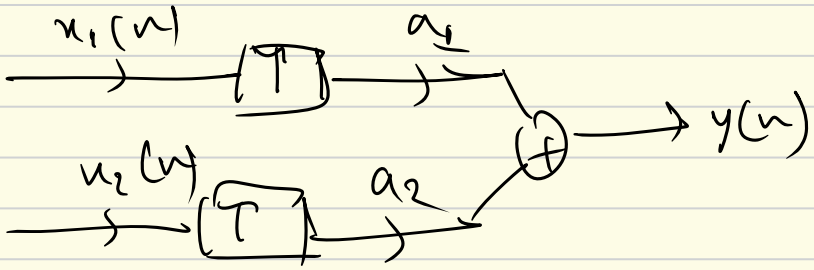
affect all variables

affect only n **

Graphical rep. of linear/nonlinear:



equal to

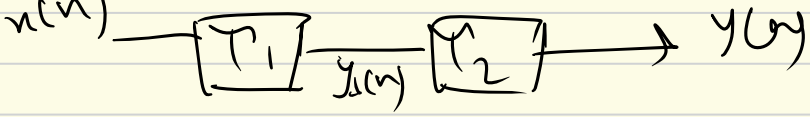


$$T[a_1 x_1(n) + a_2 x_2(n)] = a_1 T[x_1(n)] + a_2 T[x_2(n)]$$

- If system is linear and time invariant, then its input output is also time invariant and

$$T_1 T_2 = T_2 T_1$$

cascading



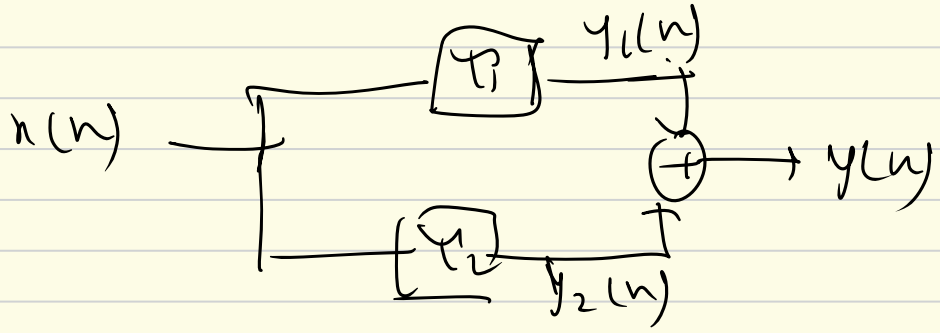
$$\therefore y_1(n) = T_1[x(n)]$$

$$\therefore y(n) = T_2[y_1(n)]$$

$$= T_2[T_1[x(n)]]$$

$$= T_2[x(n)]$$

parallel



$$y(n) = y_1(n) + y_2(n)$$

$$= T_1[x(n)] + T_2[x(n)]$$

$$= (T_1 + T_2)[x(n)]$$

$$= T_p[x(n)]$$

Convolution: An operation that combines two signals to produce a third signal. It shows how one signal modifies/filters another.

Time invariance: property of a system where a time shift in the input causes an identical time shift in the output.

$$T[x(n-k)] : y(n-k)$$

- for cascading systems: impulse responses are convoluted
- for parallel systems: impulse responses are added.

- DTFT diagram components:
  - delay element z^{-1}
  - multiplier $\times$
  - adder/summation $\oplus$ or Σ

Practice

Chapter 2:

Figure P2.2

(a) $x(n-2)$ **(b)** $x(4-n)$ **(c)** $x(n+2)$ **(d)** $x(n)u(2-n)$ **(e)** $x(n-1)\delta(n-3)$
(f) $x(n^2)$ **(g)** even part of $x(n)$ **(h)** odd part of $x(n)$

(e) $x(n) = \{1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}\}$

$x(n-1) = \{\frac{1}{\uparrow}, 1, 1, 1, \frac{1}{2}, \frac{1}{2}\}$

$\delta(n-3) = \{\underset{\uparrow}{0}, 0, 0, 1, 0, \text{—} \text{—}$

$\therefore x(n-1)\,\delta(n-3) = \{0, 0, 0, \underset{\uparrow}{1}, 0, 0 \text{—}\}$

(d) $x(n)\,u(2-n)$

$u(2-n) = \begin{cases} 2; & n \le 2 \\ 0; & n > 2 \end{cases}$

$\therefore$ output $= \{1, 1, \underset{\uparrow}{1}, 1, 0, 0 \cdots\}$

(f) $x(n^2) = \{0, x(1), x(1), x(0), x(1), x(4), x(9), x(16), \text{—}\}$

$= \{0, \frac{1}{2}, 1, \underset{\uparrow}{1}, 1, \frac{1}{2}, 0, 0 \cdots\}$

(g) even part $= \dfrac{x(n) + x(-n)}{2}$

$x(n) = \{\frac{1}{2}, \frac{1}{2}, 1, 1, \underset{\uparrow}{\frac{1}{2}}, 1\}$

$x(n) = \{0, 0, 0, -1, \underset{\uparrow}{-\frac{1}{2}}, -1, 1, \frac{1}{2}, \frac{1}{2}\}$

$x(-n) = \{\frac{1}{2}, \frac{1}{2}, 1, -1, \frac{1}{2}, -1, 0, 0, 0\}$

$x_e = \{\frac{1}{4}, \frac{1}{4}, -\frac{1}{2}, -1, 1, 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}\}$

circular symmetry as well

2.6 Consider the system

$$y(n) = \mathcal{T}[x(n)] = x(n^2)$$

(a) Determine if the system is time invariant.

(b) To clarify the result in part (a) assume that the signal

$$x(n) = \begin{cases} 1, & 0 \le n \le 3 \\ 0, & \text{elsewhere} \end{cases}$$

is applied into the system.

(1) Sketch the signal $x(n)$.

(2) Determine and sketch the signal $y(n) = \mathcal{T}[x(n)]$.

(3) Sketch the signal $y_2'(n) = y(n-2)$.

(4) Determine and sketch the signal $x_2(n) = x(n-2)$.

(5) Determine and sketch the signal $y_2(n) = \mathcal{T}[x_2(n)]$.

(6) Compare the signals $y_2(n)$ and $y(n-2)$. What is your conclusion?

(c) Repeat part (b) for the system

$$y(n) = x(n) - x(n-1)$$

Can you use this result to make any statement about the time invariance of this system? Why?

(d) Repeat parts (b) and (c) for the system

$$y(n) = \mathcal{T}[x(n)] = nx(n)$$

(a) $x(n') = x(n)$

$\therefore x(n-k) = x[(n-k)^2]$

$= x(n^2 + 2nk - k^2)$

which is non linear with respect to time and simply not a linear shift.

(b) $x(n) = \{0, \underset{\uparrow}{1}, 1, 1, 1\}$

$y(n) = \{0, x(-2), x(-1), x(0), x(1), x(2), x(3)\}$

$= \{0, 0, 1, \underset{\uparrow}{1}, 1, 0, 0\}$

not time invariant

(4) $x(n) = \{0, \underset{\uparrow}{1}, 1, 1, 1, 0\}$

$x(n-1) = \{\underset{\uparrow}{0}, 1, 1, 1, 1\}$

$\therefore y(n) = \{0, 1, 0, 0, 0, -1\}$

2.16 **(a)** If $y(n) = x(n) * h(n)$, show that $\sum_y = \sum_x \sum_h$, where $\sum_x = \sum_{n=-\infty}^{\infty} x(n)$.

(b) Compute the convolution $y(n) = x(n) * h(n)$ of the following signals and check the correctness of the results by using the test in (a).

(1) $x(n) = \{1, 2, 4\}, h(n) = \{1, 1, 1, 1, 1\}$

(2) $x(n) = \{1, 2, -1\}, h(n) = x(n)$

(3) $x(n) = \{0, 1, -2, 3, -4\}, h(n) = \{\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}\}$

(4) $x(n) = \{1, 2, 3, 4, 5\}, h(n) = \{1\}$

(5) $x(n) = \{\underset{\uparrow}{1}, -2, 3\}, h(n) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}$

(6) $x(n) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}, h(n) = \{1, \underset{\uparrow}{-2}, 3\}$

(7) $x(n) = \{\underset{\uparrow}{0}, 1, 4, -3\}, h(n) = \{\underset{\uparrow}{1}, 0, -1, -1\}$

(8) $x(n) = \{\underset{\uparrow}{1}, 1, 2\}, h(n) = u(n)$

(9) $x(n) = \{1, 1, \underset{\uparrow}{0}, 1, 1\}, h(n) = \{1, -2, -3, \underset{\uparrow}{4}\}$

(10) $x(n) = \{1, 2, \underset{\uparrow}{0}, 2, 1\} h(n) = x(n)$

(11) $x(n) = (\frac{1}{2})^n u(n), h(n) = (\frac{1}{4})^n u(n)$

(a) $y(n) = \sum_{k} h(k)\, u(n-k)$

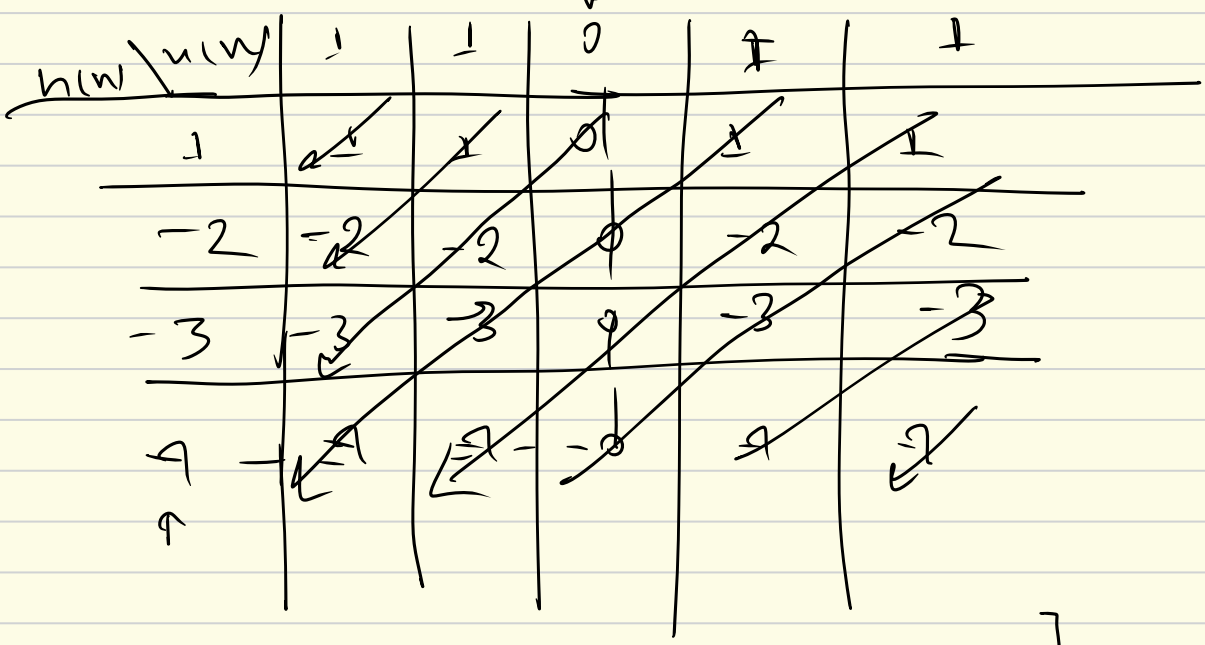
$$\therefore \sum_{n} y(n) = \sum_{n}\sum_{k} h(k)\, u(n-k)$$

$$= \sum_{k} h(k) \sum_{n} u(n-k)$$

$$= \sum h(k) \sum u(n)$$

(b) (9) $u(n) = \{1, 1, \underset{\uparrow}{0}, 1, 1\}$

$h(n) = \{1, -2, -3, \underset{\uparrow}{4}\}$



$y(n) = \{1, -1, -5, 2, 3, \underset{\uparrow}{-5}, 1, 4\}$

(11) $x(n) = \left(\frac{1}{2}\right)^n u(n)$

$h(n) = \left(\frac{1}{a}\right)^n u(n)$

$\therefore y(n) = \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k u(k) \left(\frac{1}{a}\right)^{n-k} u(n-k)$

2.21 Compute the convolution $y(n) = x(n) * h(n)$ of the following pairs of signals.

(a) $x(n) = a^n u(n)$, $h(n) = b^n u(n)$ when $a \neq b$ and when $a = b$

(b) $x(n) = \begin{cases} 1, & n = -2, 0, 1 \\ 2, & n = -1 \\ 0, & \text{elsewhere} \end{cases}$ $\quad h(n) = \delta(n) - \delta(n-1) + \delta(n-4) + \delta(n-5)$

(c) $x(n) = u(n+1) - u(n-4) - \delta(n-5)$; $h(n) = [u(n+2) - u(n-3)] \cdot (3 - |n|)$

(d) $x(n) = u(n) - u(n-5)$; $h(n) = u(n-2) - u(n-8) + u(n-11) - u(n-17)$

(a) $y(n) = \sum_k^n a^k u(k) \cdot b^{n-k} u(n-k)$

$\therefore$ if $a \neq b \Rightarrow \sum_k^n a^k b^{n-k}$

$= b^n \sum \frac{a^k}{b^k}$ which is a geometric series

$\sum r^n = \frac{1 - r^{n+1}}{1 - r}$

$\therefore y(n) = b^n \cdot \frac{1 - \left(\frac{a}{b}\right)^{n+1}}{1 - \frac{a}{b}}$

$= b^n \cdot \frac{1 - \frac{a^{k+1}}{b^{n+1}}}{\frac{b-a}{b}}$

$= b^n \times \frac{b}{b-a} \times \frac{b^{n+1} - a^{n+1}}{b^{n+1}}$

if $a = b$: $\sum a^k a^{n-k}$

$= a^n u(n)$

$\Rightarrow a^n (n+1)$

2.34 Consider a system with impulse response

$$h(n) = \begin{cases} (\frac{1}{2})^n, & 0 \le n \le 4 \\ 0, & \text{elsewhere} \end{cases}$$

Determine the input $x(n)$ for $0 \le n \le 8$ that will generate the output sequence

$$y(n) = \{\underset{\uparrow}{1}, 2, 2.5, 3, 3, 3, 2, 1, 0, \ldots\}$$

$$h = \{\underset{\uparrow}{1}, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$$

$$y(n) = \{\underset{\uparrow}{1}, 2, 2.5, 3, 3, 3, 2, 1, 0\}$$

$\therefore y[0] = h[0]\, x[0]$

$\Rightarrow 1 = 1 \times x[0]$

$\Rightarrow x[0] = 1$

$y[1] = x[0]\, h[1] + x[1]\, h[0]$

$\Rightarrow 2 = 1 \times \frac{1}{2} + x[1] \times 1$

$\Rightarrow x[1] = \frac{3}{2}$

$y[2] = x[0]\, h[2] + x[1]\, h[1] + x[2]\, h[0]$

$\Rightarrow 2.5 = 1 \times \frac{1}{4} + \frac{3}{2} \times \frac{1}{2} + x[2] \times 1$

$\Rightarrow x[2] = \frac{3}{2}$

2.35 Consider the interconnection of LTI systems as shown in Fig. P2.35.

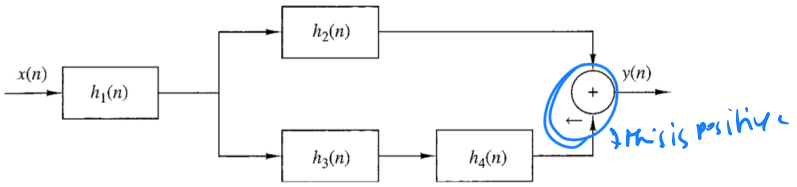


Figure P2.35

(a) Express the overall impulse response in terms of $h_1(n)$, $h_2(n)$, $h_3(n)$, and $h_4(n)$.

(b) Determine $h(n)$ when

$$h_1(n) = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\}$$

$$h_2(n) = h_3(n) = (n+1)u(n)$$

$$h_4(n) = \delta(n-2)$$

(c) Determine the response of the system in part (b) if

$$x(n) = \delta(n+2) + 3\delta(n-1) - 4\delta(n-3)$$

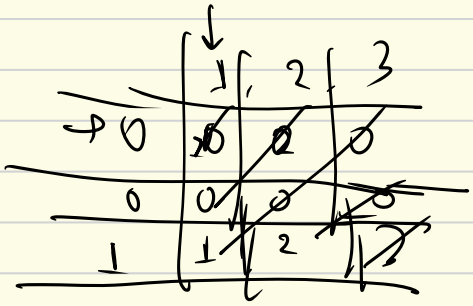
(a) $h(n) = h_1(n) * [h_2(n) + h_3(n) * h_4(n)]$

conv

conv

(b) $h_2(n) = \{0, 0, \underset{\uparrow}{1}, 2, 3\}$

$h_4(n) = \{0, 0, \underset{\uparrow}{0}, 1, 0, 0, \dots\}$



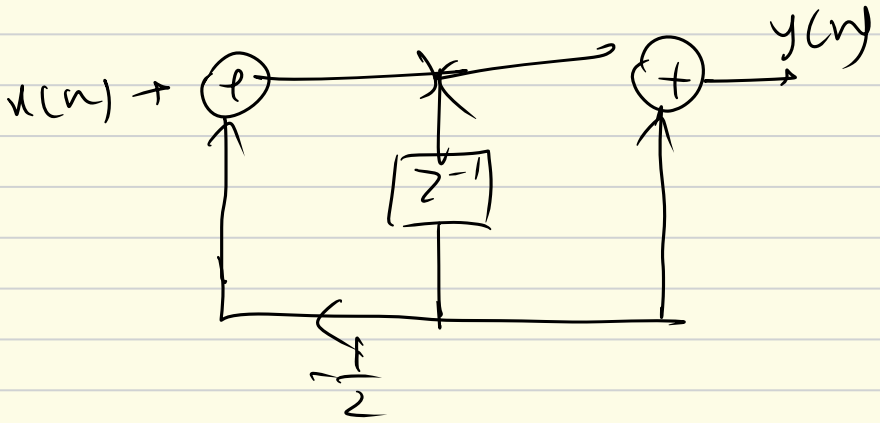
$\therefore h_3(n) * h_4(n) = \{0, 0, \underset{\uparrow}{1}, 2, 3\}$

$u(n-2) = \{0, 0, \underset{\uparrow}{1}, 1, 1\}$

$(n-1)u(n-2) = \{0, 0, \underset{\uparrow}{1}, 2, 3\}$

$\therefore h_3(n) * h_4(n) = (n-1)u(n-2)$

2.47:



$$\therefore y(n) = \frac{1}{2} y(n-1) + x(n) + x(n-1)$$

Chapter 1:

1.4 **(a)** Show that the fundamental period N_p of the signals

$$s_k(n) = e^{j2\pi kn/N}, \qquad k = 0, 1, 2, \ldots$$

is given by $N_p = N/\text{GCD}(k, N)$, where GCD is the greatest common divisor of k and N.

(b) What is the fundamental period of this set for $N = 7$?

(c) What is it for $N = 16$?

(a) Given $e^{jwn} = e^{j\frac{2\pi k}{N}n}$

$\therefore w = \frac{2\pi k}{N}$

$\Rightarrow f = \frac{k}{N}$

assuming α = GCD

$k = k'\alpha$

$N = N'\alpha$ **

as $f = \frac{k'}{N'}$ $\therefore N' = \frac{N}{\alpha}$

1.5 Consider the following analog sinusoidal signal:

$$x_a(t) = 3\sin(100\pi t)$$

(a) Sketch the signal $x_a(t)$ for $0 \le t \le 30$ ms.

(b) The signal $x_a(t)$ is sampled with a sampling rate $F_s = 300$ samples/s. Determine the frequency of the discrete-time signal $x(n) = x_a(nT)$, $T = 1/F_s$, and show that it is periodic.

(c) Compute the sample values in one period of $x(n)$. Sketch $x(n)$ on the same diagram with $x_a(t)$. What is the period of the discrete-time signal in milliseconds?

(d) Can you find a sampling rate F_s such that the signal $x(n)$ reaches its peak value of 3? What is the minimum F_s suitable for this task?

(d) $x(n) = 3$

$= 3\sin\left(\frac{100\pi k}{F_s}\right)$

$= 3\sin\left(\frac{100\pi}{F_s}\right)$

$\therefore \frac{100\pi}{F_s} = \frac{\pi}{2}$

$\therefore F_s = 200$

(1) $f = \frac{1}{6}$

$\therefore x_1(t) = 3\sin\left(2\pi \times \frac{1}{6} \times t\right)$

• $f = \frac{T}{T_p} = \frac{k}{N}$ · always divide by F_s in discrete

1.6 A continuous-time sinusoid $x_a(t)$ with fundamental period $T_p = 1/F_0$ is sampled at a rate $F_s = 1/T$ to produce a discrete-time sinusoid $x(n) = x_a(nT)$.

(a) Show that $x(n)$ is periodic if $T/T_p = k/N$ (i.e., T/T_p is a rational number).

(b) If $x(n)$ is periodic, what is its fundamental period T_p in seconds?

(c) Explain the statement: $x(n)$ is periodic if its fundamental period T_p, in seconds, is equal to an integer number of periods of $x_a(t)$.

1.6

(a)

$$\begin{aligned} x(n) &= A\cos(2\pi F_0 n/F_s + \theta) \\ &= A\cos(2\pi(T/T_p)n + \theta) \end{aligned}$$

But $T/T_p = f \Rightarrow x(n)$ is periodic if f is rational.

(b) If x(n) is periodic, then f=k/N where N is the period. Then,

$T_d = NT$ ← $T_d = (\frac{k}{f}T) = k(\frac{T_p}{T})T = kT_p.$

Thus, it takes k periods (kT_p) of the analog signal to make 1 period (T_d) of the discrete signal.

(c) $T_d = kT_p \Rightarrow NT = kT_p \Rightarrow f = k/N = T/T_p \Rightarrow$ f is rational $\Rightarrow$ x(n) is periodic.

* during D/A conversion, we multiply using F_s

1.10 A digital communication link carries binary-coded words representing samples of an input signal

$$x_a(t) = 3\cos 600\pi t + 2\cos 1800\pi t$$

The link is operated at 10,000 bits/s and each input sample is quantized into 1024 different voltage levels.

(a) What are the sampling frequency and the folding frequency?

(b) What is the Nyquist rate for the signal $x_a(t)$?

(c) What are the frequencies in the resulting discrete-time signal $x(n)$?

(d) What is the resolution Δ?

(a) $b = \log_2 1024 = 10$

$\therefore$ sampling frequency $= \frac{10,000}{10}$

$= 1000$ samples/sec

$\therefore$ $f_{fold} = 500$ Hz

(c) $\frac{1800}{200} = 9 \Rightarrow 900 \therefore 0.9 > 1.5$ sampling

$\therefore = 0.4$

(d) $A = \frac{5-(-5)}{1023} =$

$(L-1)$

1.11 Consider the simple signal processing system shown in Fig. P1.11. The sampling periods of the A/D and D/A converters are $T = 5$ ms and $T' = 1$ ms, respectively. Determine the output $y_a(t)$ of the system, if the input is

$$x_a(t) = 3\cos 100\pi t + 2\sin 250\pi t \qquad (t \text{ in seconds})$$

The postfilter removes any frequency component above $F_s/2$.

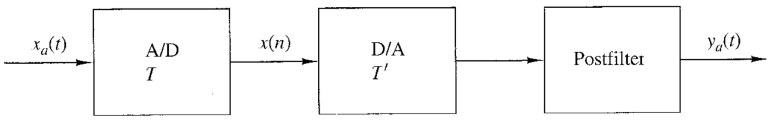


Figure P1.11

Given $T = 5\text{ms} = 5 \times 10^{-3}$

$\therefore F = \frac{1}{T} = \frac{1}{5 \times 10^{-3}} = 200$ sampling

$\therefore 3\cos\left(\frac{\pi n}{2}\right) - 2\sin\left(\frac{3\pi n}{4}\right)$

Now, for discrete to analog

$t = \frac{n}{F_s}$

$\therefore n = F_s \times t$

Since D/A $F_s = 1\text{ms} = 1000\left(\frac{1}{T}\right)$

$\therefore 3\cos\left(\frac{\pi n}{2}\right) \Rightarrow 3\cos\left(\frac{\pi \times 1000t}{2}\right)$

1.14 Determine the bit rate and the resolution in the sampling of a seismic signal with dynamic range of 1 volt if the sampling rate is $F_s = 20$ samples/s and we use an 8-bit A/D converter. What is the maximum frequency that can be present in the resulting digital seismic signal?

bit rate = sample number × bit in each

$= 20 \times 8$

$= 160$

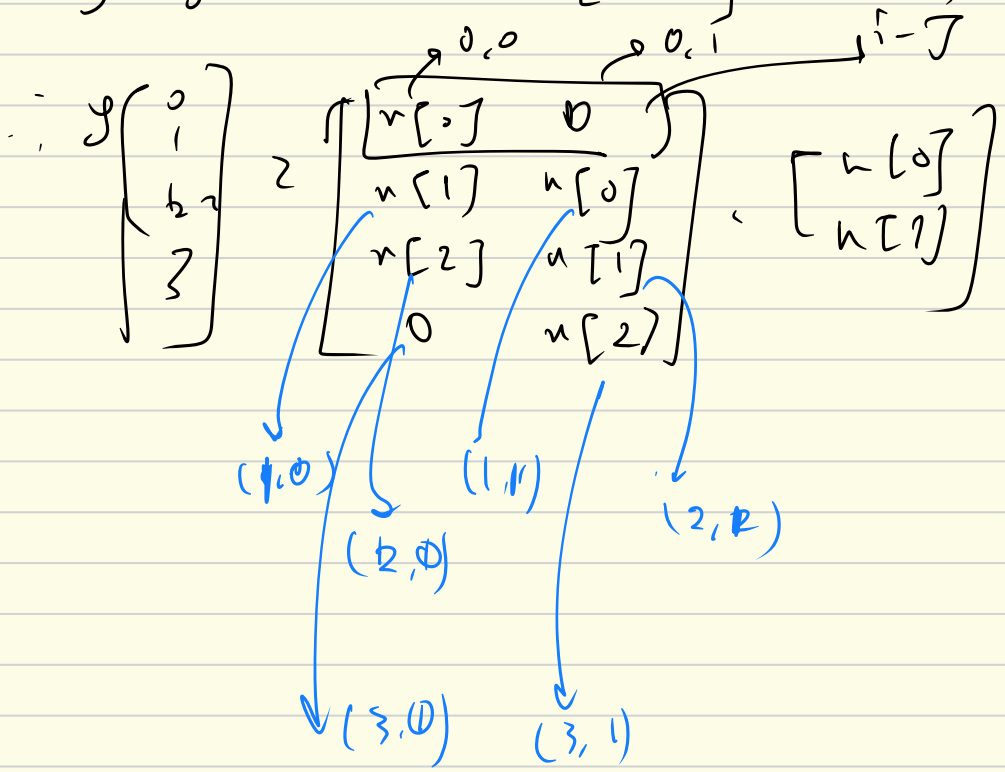
resolution $= \frac{\text{volt range}}{L-1} = \frac{1 \text{ volt}}{2^8 - 1}$

f_{fold} = max frequency $= \frac{20}{2} = 10$ samples/sec

Convolution closed form:

$$y[n] = \sum_k h[k]\, x[n-k]$$

$$\therefore y\begin{bmatrix}0\\1\\2\\3\end{bmatrix} = \begin{bmatrix} x[0] & 0 \\ x[1] & x[0] \\ x[2] & x[1] \\ 0 & x[2] \end{bmatrix} \cdot \begin{bmatrix} h[0] \\ h[1] \end{bmatrix}$$



$$\therefore y = Hx$$

$$\Rightarrow H = x^{-1}y$$

Quiz-3:

- $Rex[k] = \sum_{i=0}^{n-1} x[i]\cos(2\pi ki/N)$

  $Im[k] = -\sum_{i=0}^{n-1} x[i]\sin(2\pi ki/N)$ } analysis

- $x[i] = \sum_{k=0}^{N+2} \bar{R}ex[k]\cos(2\pi ki/N) + \sum_{k=0}^{N/2} \bar{I}m\,x[k]\sin(2\pi ki/N)$ ↓ synthesis

  for all: $\bar{R}ex[k] = \frac{Rex[k]}{N/2}$

  $\bar{I}m\,x[k] = -\frac{Im\,X[k]}{N/2}$

  for 0: $\bar{R}ex[0] = \frac{Rex[0]}{N}$

  for N/2: $\bar{R}ex(N/2) = \frac{Rex(N/2)}{N}$

- $A\cos(\omega t) + B\sin(\omega t) = M\cos(\theta + \omega t)$

  $Mag\,x[k] = \sqrt{Rex[k]^2 + Im\,X[k]^2}$

  $phase[k] = \tan^{-1}\left(\frac{Im[k]}{Rex[k]}\right)$

  $Rex[k] = Mag\,X[k]\cos(Phase[k])$

  $Im\,X[k] = Mag\,X[k]\sin(Phase[k])$

- Unwrapping:

  $C_i = CINT\left(\frac{UWphase[k_{i-1}] - phase[k_i]}{2*\pi}\right)$

  $UWphase[k_i] = phase[k_i] + C_i * 2*\pi$

- $X[f] \times H[f] = Y[f]$

$\text{mag}Y[f] = \text{mag}X[f] \times \text{mag}H[f]$

$\text{phase}\,Y[f] = \text{phase}\,X[f] + \text{phase}\,H[f]$

$\text{Re}\,Y[f] = \text{Re}X[f] \times \text{Re}H[f] - \text{Im}X[f]\,\text{Im}H[f]$

$\text{Im}Y[f] = \text{Im}X[f] \times \text{Re}H[f] + \text{Re}X[f]\,\text{Im}H[f]$

- $H[f] = Y[f] / X[f]$

$\text{mag}H[f] = \text{mag}Y[f] / \text{mag}X[f]$

$\text{phase}\,H[f] = \text{phase}\,Y[f] - \text{phase}\,X[f]$

$$\text{Re}H[f] = \frac{\text{Re}X[f]\,\text{Re}Y[f] + \text{Im}X[f]\,\text{Im}Y[f]}{\text{Re}X^2[f] + \text{Im}X^2[f]}$$

$$\text{Im}H[f] = \frac{\text{Im}Y[f]\,\text{Re}X[f] - \text{Im}X[f]\,\text{Re}Y[f]}{\text{Re}X^2[f] + \text{Im}X^2[f]}$$